

ReCATS: Replay-Free Continual Anomaly Detection for Non-Stationary Multivariate Time Series

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Abstract

Multivariate time series anomaly detection is critical in safety-critical domains such as industrial monitoring and financial systems. However, real-world time series are inherently non-stationary, with evolving data distributions driven by changing operational regimes and system dynamics. As a result, most existing methods, which assume static data distributions, exhibit severe performance degradation over time and are prone to catastrophic forgetting when incrementally updated. To address these challenges, we propose *ReCATS*, a replay-free framework for continual anomaly detection in non-stationary multivariate time series. *ReCATS* combines multi-regime modeling with a dual-phase generative alignment mechanism, enabling effective adaptation to distribution shifts while preserving knowledge acquired from past tasks. Furthermore, we introduce a dynamic dual-scale thresholding strategy to mitigate decision boundary drift under evolving data distributions. Extensive experiments on six real-world benchmark datasets demonstrate that *ReCATS* consistently outperforms state-of-the-art methods in terms of detection accuracy, knowledge retention, and transferability, as evaluated by standard continual learning metrics, including Backward Transfer and Forward Transfer. The code is available at <https://github.com/Li-Qiuyang/ReCATS>.

CCS Concepts

• Computing methodologies → Anomaly detection.

Keywords

Multivariate Time Series, Anomaly Detection, Continual Learning

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1 Introduction

Multivariate time series anomaly detection (MTSAD) is a fundamental task in many safety-critical domains, including industrial system monitoring and financial risk management [2, 5, 8, 27]. In real-world scenarios, multivariate time series data streams are inherently non-stationary, where underlying data distributions—and consequently the definitions of normality—continuously evolve due to changing operational regimes and system dynamics [16]. This phenomenon, commonly referred to as *concept drift*, poses a fundamental challenge to anomaly detection systems.

Static models trained on historical data are unable to accommodate newly emerging patterns, often resulting in excessive false alarms. Conversely, naively adapting models through incremental updates, such as fine-tuning on recent data, frequently leads to *catastrophic forgetting* [13], where previously acquired knowledge is overwritten. As a result, when historical behaviors reoccur—e.g., cyclic operational regimes or seasonal patterns—the model fails to recognize them, causing severe performance degradation.

These characteristics make MTSAD a natural fit for the continual learning paradigm, which aims to incrementally adapt models to evolving data distributions while preserving previously acquired knowledge. Continual learning has been extensively studied in computer vision to address the stability-plasticity dilemma [4, 30], demonstrating strong effectiveness in continual anomaly detection under evolving environments. Inspired by these advances, extending continual learning to multivariate time series anomaly detection provides a promising direction for handling streaming data with frequent distribution shifts.

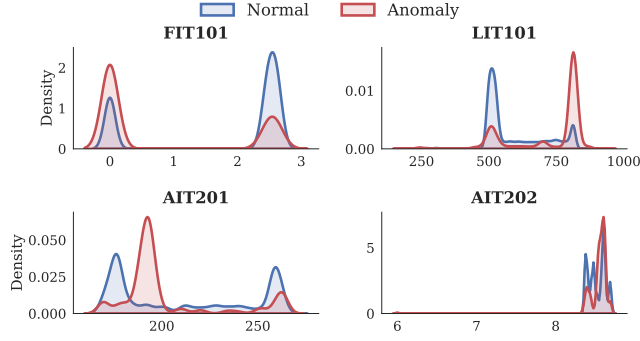


Figure 1: Probability Density Estimation of the SWaT Dataset.

To the best of our knowledge, GenDeX [9] is the first work to introduce generative replay into continual multivariate time series anomaly detection. While it demonstrates the feasibility of using synthetic data to alleviate catastrophic forgetting, GenDeX primarily emphasizes data generation quality. Moreover, it lacks a systematic evaluation of anomaly detection performance across sequential tasks using standard continual learning metrics.

To bridge these gaps, we propose **ReCATS**, a framework for Replay-free Continual Anomaly detection for non-stationary multivariate Time Series, designed to address the following challenges. In this work, “replay-free” refers to continual learning without storing historical raw data. Instead, historical knowledge is retained through generated pseudo-samples.

Challenge I: Distribution complexity across multiple operational regimes. Real-world time series data often exhibit complex, multi-modal distributions arising from multiple operational regimes. These distributions evolve over time and frequently exhibit substantial overlap between normal and anomalous patterns, making it difficult to distinguish genuine anomalies.

Example 1.1. As illustrated in Figure 1, the empirical distributions of representative sensor variables from the SWaT dataset (e.g., FIT101, LIT101, AIT201, and AIT202) exhibit pronounced multi-peaked structures under both normal and anomalous conditions. These complex and heterogeneous patterns reflect the presence of multiple operational regimes within the system. At the same time, substantial overlap is observed between the distributions of normal and anomalous samples across these variables, indicating that anomalies are embedded within a heterogeneous background of normal behaviors rather than manifesting as simple global shifts. Such distributional complexity makes it fundamentally challenging to distinguish genuine anomalies from regime transitions.

Challenge II: Severe catastrophic forgetting under continual learning. In the presence of continuously evolving data distributions, models trained in an incremental manner are particularly vulnerable to catastrophic forgetting when historical data are no longer accessible. Without effective mechanisms to preserve previously acquired knowledge, performance on earlier tasks can deteriorate rapidly, undermining the stability and reliability of anomaly detection across long task sequences.

Challenge III: Limited adaptability in anomaly discrimination. Anomaly detection performance in time series critically depends on threshold selection. However, fixed thresholds are often inadequate due to substantial distributional shifts across tasks and

non-stationary fluctuations within individual tasks. Global thresholds tend to overlook local dynamics, leading to reduced sensitivity, whereas purely local thresholds are highly susceptible to noise and extreme values. This trade-off results in unstable detection behavior, highlighting the need for more adaptive and robust anomaly discrimination strategies in continual settings.

To address these challenges, **ReCATS** integrates *multi-regime modeling* to capture complex distributional patterns across operational states, together with a *dual-phase generative alignment mechanism* to mitigate catastrophic forgetting in replay-free continual learning. In addition, a *dynamic dual-scale thresholding strategy* is introduced to enable adaptive and robust anomaly discrimination under evolving data distributions.

Our main contributions are summarized as follows.

- We propose **ReCATS**, a unified framework for replay-free continual anomaly detection that explicitly addresses multi-regime distribution complexity and catastrophic forgetting in non-stationary multivariate time series.
- We introduce a replay-free incremental learning strategy by combining multi-regime modeling with a dual-phase generative alignment mechanism, enabling effective knowledge retention across sequential tasks.
- We design a dynamic dual-scale thresholding strategy to adaptively balance sensitivity and robustness in non-stationary time series anomaly detection.
- Extensive experiments on six real-world benchmark datasets demonstrate that ReCATS consistently achieves superior detection accuracy and strong knowledge retention, as measured by standard continual learning metrics including BWT and FWT.

2 Related Work

Static time series anomaly detection. Traditional methods assume static data distributions and are broadly categorized into machine learning and deep learning approaches [23]. Classical techniques such as LOF [3], One-Class SVM [24], k-means [11], and Isolation Forest [14] struggle with high-dimensional temporal dependencies. Deep reconstruction models including AE [22], VAE [20], Omni-Anomaly, InterFusion [28], and BeatGAN [36] learn latent representations. Recent innovations involve diffusion-based D3R [29] and Transformers such as Anomaly Transformer [32] and DCdetector [33], focusing on association discrepancies and local patches.

Prediction-based methods leverage temporal autocorrelation [19], typically using LSTM networks [34], but face limited context windows and performance decline under concept drift. Importantly, all static approaches assume fixed data distributions. When applied to evolving streams, they suffer from catastrophic forgetting. Robust variants handling non-stationarity remain designed for single-task scenarios and lack incremental update mechanisms without full retraining.

Continual learning and its scarcity in time series. Continual learning aims to reconcile plasticity for new tasks with stability for historical knowledge. Although extensively studied in image classification [12, 35], its application to anomaly detection is relatively recent. Most existing frameworks focus on pixel level image anomalies. These methods typically do not account for the unique temporal

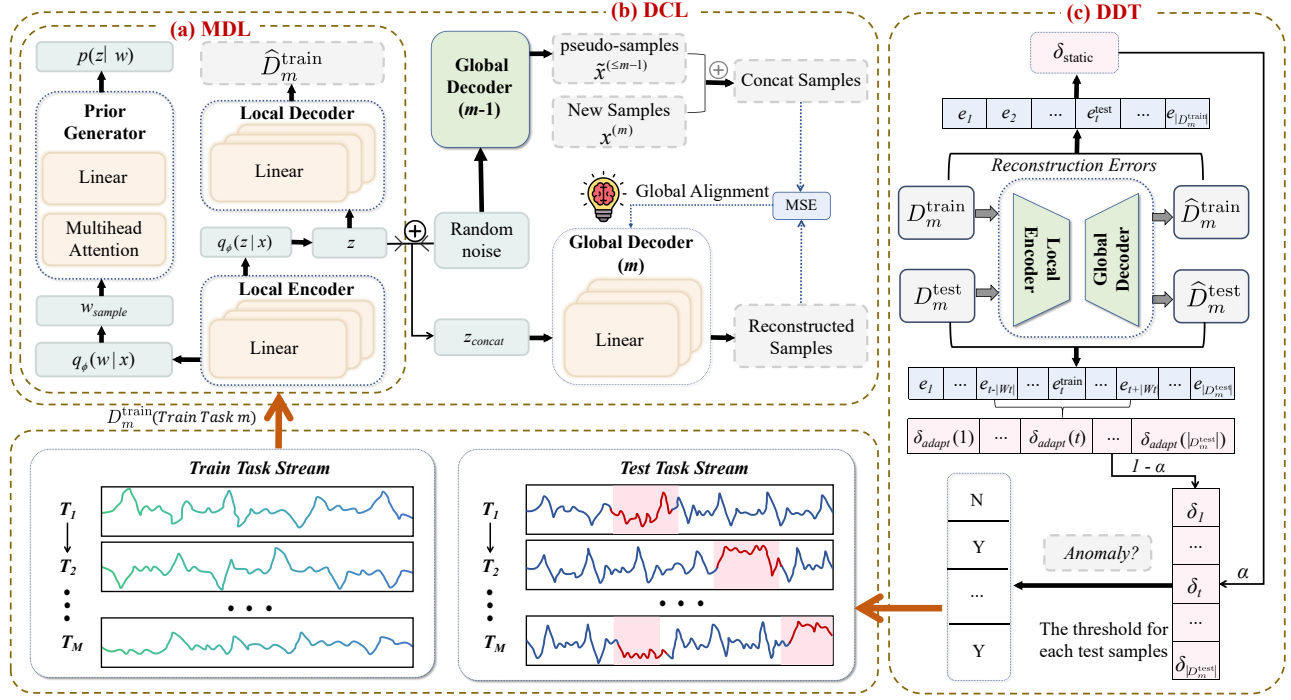


Figure 2: Overall Framework of ReCATS

dependencies and sparse anomaly distributions inherent in multivariate time series which limits their effectiveness in non-stationary sequential environments. To our knowledge, GenDex [9] is the only prior research that introduced a generative replay framework for this task. However, GenDex primarily validates the feasibility of using synthetic data under a unimodal Gaussian assumption. ReCATS extends beyond mere reconstruction feasibility by providing the first systematic evaluation of knowledge retention and adaptation using task level metrics.

In this paper, we study continual anomaly detection for multivariate time series (continual MTSAD), where data distributions evolve over time and models must incrementally learn from a sequence of tasks without access to raw historical data.

In this setting, the objective is to develop a generative anomaly detection model that can continuously adapt to newly arriving data while preserving anomaly detection performance on previously learned tasks. Specifically, the model is trained following a temporally ordered stream of tasks. After completing training on each task, it is expected to maintain effective anomaly detection performance on both the current task and all previously encountered tasks, despite distribution shifts across tasks and the absence of historical training data.

3 Preliminaries

Definition 3.1 (Multivariate Time Series (MTS)). A sequence of observations collected over time from multiple correlated variables. Formally, given a multivariate time series of length T and dimensionality d , it is denoted as

$$\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_T), \quad \mathbf{X}_t \in \mathbb{R}^d,$$

where $\mathbf{X}_t$ represents the d -dimensional observation at time step t .

Definition 3.2 (Task Stream). In the continual learning setting, the entire time series stream is chronologically partitioned into M consecutive and non-overlapping segments, each corresponding to an anomaly detection task. The resulting task stream is defined as

$$\mathcal{T} = \{\mathcal{T}_1, \mathcal{T}_2, \dots, \mathcal{T}_M\}.$$

Each task $\mathcal{T}_m$ is associated with a dataset $\mathcal{D}_m$, which consists of a training set $\mathcal{D}_m^{\text{train}}$ containing only normal data and a test set $\mathcal{D}_m^{\text{test}}$ that includes both normal samples and anomalies.

Problem Statement. Given a temporally ordered task stream $\mathcal{T} = \{\mathcal{T}_1, \dots, \mathcal{T}_M\}$ derived from an evolving multivariate time series, the goal of continual MTSAD is to incrementally learn anomaly detection models from a sequence of tasks without access to raw data from previous tasks. After training on task $\mathcal{T}_m$, the model is required to effectively detect anomalies in the test sets of all tasks $\{\mathcal{T}_1, \dots, \mathcal{T}_m\}$, despite distribution shifts across tasks.

4 Framework

4.1 Framework Overview

As illustrated in Figure 2, **ReCATS** is composed of three functional modules that jointly address the key challenges of continual anomaly detection in multivariate time series: (i) a **multi-regime distribution learning** (MDL) module, (ii) a **dual-phase continual learning** (DCL) module, and (iii) a **dynamic dual-scale thresholding** (DDT) module.

- **MDL Module:** is designed to model evolving data distributions arising from multiple operational regimes, enabling the framework to capture complex and heterogeneous distributional patterns commonly observed in real-world multivariate time series.
- **DCL Module:** facilitates replay-free continual learning by balancing task-specific adaptation and long-term knowledge retention through a dual-phase training mechanism.
- **DDT Module:** adaptively determines anomaly boundaries using complementary static and adaptive thresholds to enhance robustness under non-stationary conditions.

Overall, the MDL module encodes the input multivariate time series x into latent representations z , which are utilized by the DCL and DDT modules for continual generative modeling and reconstruction-based anomaly detection, respectively. During training, MDL learns task-specific representations from current data, while DCL maintains a global decoder through generative replay using current and generated pseudo-samples, thereby mitigating catastrophic forgetting. During inference, the MDL encoder and DCL decoder jointly reconstruct the input sequence $\hat{x}$, and DDT identifies anomalies based on reconstruction errors with adaptive thresholds.

These modules jointly address distributional complexity (*Challenge I*), continual adaptation (*Challenge II*), and robust anomaly decision-making (*Challenge III*) in non-stationary environments.

4.2 MDL Module

The MDL module serves as the generative backbone of ReCATS, aiming to model the complex and evolving data distributions induced by multiple operational regimes in non-stationary multivariate time series. By constructing a structured latent representation, it provides robust reconstruction capability while enabling regime-aware anomaly discrimination.

Backbone Architecture. In a continual learning setting, each incoming task $\mathcal{T}_m \in \mathcal{T}$ typically contains data generated under multiple operational regimes, rather than a single homogeneous distribution. Such intra-task distributional heterogeneity poses a fundamental challenge for anomaly detection, as regime transitions may closely resemble anomalous behaviors. To explicitly capture this complexity, the MDL module projects each input sample x from the current training set $\mathcal{D}_m^{\text{train}}$ into a structured latent space.

Specifically, the MDL module adopts an encoder-decoder architecture, where the encoder maps input samples into disentangled latent variables and the decoder reconstructs the original observations from these representations. Instead of performing simple dimensionality reduction, the encoder decomposes the underlying generative factors into three complementary components: a continuous latent variable z that captures fine-grained sample characteristics for reconstruction, a discrete latent variable c that encodes latent operational regimes, and a contextual latent variable w that summarizes temporal and contextual information. Based on this design, the encoder infers the following posterior distributions:

$$q_\phi(z|x) = \mathcal{N}(z|\mu_z(x), \text{diag}(\sigma_z^2(x))), \quad (1)$$

$$q_\phi(w|x) = \mathcal{N}(w|\mu_w(x), \text{diag}(\sigma_w^2(x))), \quad (2)$$

$$q_\phi(c|x) = \text{Cat}(c|\pi(x)), \quad c \in \{1, \dots, K\}. \quad (3)$$

Here, $\pi(x)$ denotes a K -dimensional probability vector, where each element represents the likelihood that sample x originates from the k -th operational regime. To enable end-to-end differentiable training, we employ the standard reparameterization trick for continuous latent variables, i.e., $z = \mu_z(x) + \sigma_z(x) \odot \epsilon$, with $\epsilon \sim \mathcal{N}(0, I)$. **Conditional Prior Generator.** While the structured latent space captures intra-task heterogeneity, a fixed prior distribution is insufficient to accommodate evolving operational regimes under non-stationary conditions. To address this limitation, we introduce a conditional prior generator that dynamically parameterizes a Gaussian mixture prior for z conditioned on the contextual variable w .

The generator leverages a multi-head attention mechanism to model complex dependencies within the contextual representation and to emphasize salient temporal features. Based on the attention-enhanced features, independent fully connected layers produce the mean $\mu^{(k)}(w)$ and variance $\sigma^{(k)}(w)$ for each mixture component $k \in \{1, \dots, K\}$. This results in the following conditional Gaussian mixture prior:

$$p(z|w) = \frac{1}{K} \sum_{k=1}^K \mathcal{N}(z|\mu^{(k)}(w), \text{diag}(\sigma^{(k)}(w)^2)). \quad (4)$$

By conditioning the latent prior on w , the model can dynamically adapt its latent space to reflect distributional shifts induced by changing operational regimes. During decoding, the latent variable z is mapped back to the input space to reconstruct the original observation $\hat{x}$. The local training objective combines reconstruction fidelity with KL regularization to enforce structured latent representations:

$$\begin{aligned} \mathcal{L}_{\text{local}} = & \|x - \hat{x}\|^2 + \text{KL}(q_\phi(w|x) \| p(w)) \\ & + \text{KL}(q_\phi(c|x) \| p(c)) \\ & + \mathbb{E}_{q_\phi(c,x)} [\text{KL}(q_\phi(z|x) \| p_\theta(z|w, c))]. \end{aligned} \quad (5)$$

The continuous latent variable w follows a standard Gaussian prior, $p(w) = \mathcal{N}(0, I)$, while the discrete latent variable c follows a categorical prior, $p(c) = \text{Cat}(\pi)$ with $\pi_k = \frac{1}{K}$, where K denotes the number of Gaussian mixture components.

This design enables the MDL module to effectively capture non-stationary and multi-regime characteristics of time series data, thereby establishing a robust generative foundation for subsequent replay-free continual learning and dynamic thresholding.

4.3 DCL Module

The DCL module enables continual learning in ReCATS by balancing task-specific adaptation and long-term knowledge retention through a dual-phase training strategy. By explicitly decoupling fast local adaptation from global knowledge consolidation, it mitigates catastrophic forgetting in replay-free settings without storing raw historical data.

Local Training Phase. When a new task arrives, the model first performs a local training phase using only data from the current task. This phase is designed to rapidly adapt the model to task-specific distributional changes in the evolving time series, prioritizing plasticity and fast learning. During local training, task-relevant components are updated to capture newly emerging patterns, enabling effective adaptation to distribution shifts.

For the initial task $\mathcal{T}_1$, both the encoder and decoder are jointly optimized to establish a stable generative representation. The resulting decoder is then used to initialize the global decoder. For subsequent tasks, local training updates only the encoder and task-specific components, while the global decoder is kept fixed at this stage. This design allows efficient adaptation to new tasks while avoiding unnecessary disruption of previously learned generative structures.

Global Alignment Phase. Although local training facilitates rapid task adaptation, it is insufficient to prevent catastrophic forgetting under continual learning. To address this issue, a global alignment phase is introduced after local training to consolidate knowledge across tasks. Under the replay-free constraint, this phase leverages a generative replay strategy to align representations from historical and current tasks within a unified generative model.

Specifically, the global decoder obtained from the previous task is frozen and used to generate pseudo-samples that approximate the data distributions of previously learned tasks. These pseudo-samples are combined with real samples from the current task to form a mixed training set, which is then used to update the global decoder. This process enables effective knowledge retention without storing raw historical data, making it suitable for long-term deployment in streaming environments.

Formally, when task $\mathcal{T}_m$ arrives, the global decoder learned from task $\mathcal{T}_{m-1}$ generates pseudo-samples $\tilde{x}^{(\leq m-1)}$ for historical tasks. The global decoder $g(\cdot)$ is optimized by minimizing:

$$\begin{aligned} \mathcal{L}_{\text{global}} = & \mathbb{E}_{z^{(m)}} [\|x^{(m)} - g(z^{(m)})\|^2] \\ & + \mathbb{E}_{z^{(\leq m-1)}} [\|\tilde{x}^{(\leq m-1)} - g(z^{(\leq m-1)})\|^2], \end{aligned} \quad (6)$$

where the first term encourages accurate modeling of the current task, and the second term preserves knowledge from historical tasks through pseudo-sample replay.

By decoupling fast local adaptation from global knowledge alignment, the DCL module effectively addresses catastrophic forgetting while maintaining cross-task consistency and long-term stability in continual time-series anomaly detection.

4.4 DDT Module

The DDT module is responsible for translating reconstruction errors into reliable anomaly decisions under non-stationary data distributions. Although reconstruction-based models provide informative anomaly scores, evolving data distributions often lead to decision boundary drift, rendering fixed thresholds unreliable. By integrating complementary static and adaptive thresholds, the DDT module explicitly addresses both long-term distributional shifts and short-term temporal fluctuations.

Following Definition 3.2, each task $\mathcal{T}_m \in \mathcal{T}$ corresponds to a consecutive and non-overlapping time series segment, with the associated training and test sets denoted as $\mathcal{D}_m^{\text{train}}$ and $\mathcal{D}_m^{\text{test}}$, respectively.

Static Threshold Determination. The static threshold δ_{static} is derived from reconstruction errors on the training set $\mathcal{D}_m^{\text{train}}$, serving as a task-level reference that captures the long-term statistical characteristics of the current data distribution. Specifically, it is defined as a high quantile of the empirical error distribution:

$$\delta_{\text{static}} = \text{Quantile}_\gamma(\{e_t^{\text{train}}\}_{x_t \in \mathcal{D}_m^{\text{train}}}), \quad (7)$$

where e_t^{train} denotes the reconstruction error of training sample x_t^{train} , and $\text{Quantile}_\gamma(\cdot)$ returns the γ -quantile.

Adaptive Threshold Determination. While the static threshold provides global stability, it is insufficient to capture short-term temporal variations during testing. To model local error dynamics, an adaptive threshold is computed for each test time step. Given a test sequence of length T , the adaptive threshold $\delta_{\text{adapt}}(t)$ at time step t ($1 \leq t \leq T$) is obtained by constructing a sliding window W_t centered at t over the reconstruction error sequence. After removing extreme values within W_t to improve robustness against noise and outliers, the local mean μ_t and standard deviation σ_t are computed over the filtered window, and $\delta_{\text{adapt}}(t)$ is defined as:

$$\delta_{\text{adapt}}(t) = \mu_t + 3\sigma_t. \quad (8)$$

Dual-scale Fusion. To balance long-term stability and short-term adaptivity, the final decision threshold for each test sample x_t is obtained by linearly combining the static and adaptive thresholds:

$$\delta_t = \alpha \delta_{\text{static}} + (1 - \alpha) \delta_{\text{adapt}}(t), \quad (9)$$

where $\alpha \in [0, 1]$ controls the trade-off between global consistency and local responsiveness. A test sample is classified as anomalous if its reconstruction error exceeds the fused threshold δ_t .

By integrating long-term statistical regularities with local temporal dynamics, the proposed dynamic dual-scale thresholding strategy enables robust yet responsive anomaly detection under non-stationary conditions. In addition, we adopt the widely used point adjustment strategy [25, 28, 32, 33] during evaluation to refine detection boundaries, aligning predicted anomalies more closely with true anomalous segments.

5 Experiments

We evaluate ReCATS to address the following research questions:

- **Q1:** How does ReCATS perform in terms of overall detection performance and knowledge retention across tasks compared to state-of-the-art methods?
- **Q2:** How efficient and scalable is ReCATS?
- **Q3:** How does ReCATS perform when training with different sample sizes?
- **Q4:** How sensitive is ReCATS to key hyperparameters?
- **Q5:** How do individual components contribute to the overall performance of ReCATS?

5.1 Experiment Settings

Datasets. We evaluate ReCATS on six publicly available benchmark datasets spanning diverse application domains. Specifically, Weather [6] and PSM [1] are used for meteorological and server monitoring, MSL and SMAP [10] from NASA represent space telemetry systems, and SWaT [17] together with NIPS-TS-GECCO [18] correspond to critical infrastructure and industrial control systems.

To emulate realistic non-stationary environments, each dataset is chronologically partitioned into a stream of consecutive tasks, enabling the evaluation of incremental adaptation and long-term knowledge retention under continual learning. Detailed dataset statistics and task partitioning protocols are provided in Appendix A.1.

Baselines. We compare ReCATS with six state-of-the-art baselines proposed in the last three years specifically ModernTCN [15],

Table 1: Performance comparison between ReCATS and representative baselines. Bold indicates the best.

Dataset	Methods	Accuracy				F1				ROC_AUC			
		Average	BWT	FWT	FM	Average	BWT	FWT	FM	Average	BWT	FWT	FM
Weather	ModernTCN	0.7437	-0.0005	0.7592	0.0035	0.6967	-0.0012	0.7481	0.0050	0.7820	-0.0007	0.8016	0.0033
	D3R	0.7496	0.0049	0.7660	0.0647	0.7951	-0.0028	0.8141	0.0371	0.7365	0.0057	0.7511	0.0818
	DCdetector	0.8107	0.0021	0.7997	0.0040	0.8228	0.0019	0.8174	0.0036	0.8382	0.0019	0.8333	0.0048
	ATransformer	0.8053	0.0050	0.7923	0.0003	0.8188	0.0044	0.8109	0.0003	0.8317	0.0052	0.8255	0.0007
	CLAD	0.6066	-0.0010	0.6205	0.0058	0.7523	-0.0001	0.7622	0.0003	0.5710	-0.0024	0.5763	0.0033
	DADA	0.3808	0.0056	0.38	0.0127	0.5325	0.0018	0.5295	0.0044	0.5171	0.0034	0.5182	0.0091
	ReCATS	0.8617	0.0263	0.8356	-0.0053	0.8767	0.0259	0.8530	-0.0054	0.8708	0.0240	0.8506	-0.0041
PSM	ModernTCN	0.9841	0.0044	0.9838	0.0006	0.9658	0.0094	0.9682	0.0025	0.9766	0.0076	0.9768	0.0006
	D3R	0.9798	0.0006	0.9870	0.0146	0.9193	0.0013	0.9647	0.0758	0.9465	0.0002	0.9752	0.0450
	ATransformer	0.9913	-0.0017	0.9934	0.0021	0.9706	-0.0128	0.9844	0.0135	0.9802	-0.0118	0.9917	0.0124
	DCdetector	0.9901	0.0041	0.9750	0.0043	0.9663	0.0074	0.9338	0.0195	0.9767	0.0066	0.9621	0.0172
	CLAD	0.5989	0.0031	0.5674	0.0046	0.6230	-0.0013	0.6379	0.0051	0.5867	-0.0015	0.6156	0.0095
	DADA	0.796	0.0038	0.7385	0.0368	0.6675	0.0063	0.6481	0.0922	0.7983	0.0083	0.7542	0.0209
	ReCATS	0.9930	0.0069	0.9938	-0.0038	0.9831	0.0111	0.9866	-0.0051	0.9912	0.0103	0.9920	-0.0073
MSL	ModernTCN	0.9394	-0.0025	0.9012	0.0040	0.7778	-0.0117	0.6358	0.0205	0.8655	-0.0080	0.7947	0.0143
	D3R	0.9595	-0.0012	0.9373	0.0041	0.8813	-0.0054	0.7837	0.0131	0.9478	-0.0044	0.9051	0.0065
	ATransformer	0.9694	-0.0003	0.9649	0.0009	0.9160	-0.0009	0.8984	0.0033	0.9486	-0.0003	0.9360	0.0007
	DCdetector	0.9609	-0.0012	0.9562	0.0053	0.8893	-0.0061	0.8534	0.0147	0.9384	-0.0066	0.9092	0.0110
	CLAD	0.6507	0.0000	0.7073	0.0029	0.4688	0.0000	0.4259	0.0000	0.5292	0.0000	0.5179	0.0003
	DADA	0.7501	-0.007	0.8347	0.0711	0.5418	-0.0127	0.6203	0.1473	0.8149	0.017	0.8756	0.0741
	ReCATS	0.9730	0.0001	0.9658	0.0000	0.9259	0.0002	0.9035	0.0000	0.9604	0.0001	0.9492	0.0000
NIPS-TS-GECCO	ModernTCN	0.9715	0.0005	0.9938	0.0015	0.7894	0.0000	0.9186	0.0131	0.8580	0.0021	0.9402	0.0073
	D3R	0.9646	0.0013	0.9836	0.0007	0.5663	0.0038	0.7221	0.0032	0.7874	0.0007	0.9098	0.0004
	ATransformer	0.9707	0.0076	0.9745	0.0044	0.6618	0.0341	0.5975	0.0300	0.8220	0.0230	0.8121	0.0278
	DCdetector	0.9539	0.0037	0.9436	-0.0031	0.6368	0.0192	0.2925	-0.0156	0.8210	0.0129	0.6631	-0.0121
	CLAD	0.9678	0.0003	0.9849	0.0002	0.8066	-0.0007	0.8736	0.0031	0.9066	-0.0031	0.9384	0.0040
	DADA	0.98	-0.0024	0.9953	0.0068	0.7029	-0.0583	0.7799	0.1214	0.99	-0.0012	0.996	0.0035
	ReCATS	0.9847	0.0178	0.9957	-0.0114	0.8835	0.0850	0.9574	-0.0464	0.9901	0.0474	0.9963	-0.0400
SMAP	ModernTCN	0.8672	-0.0086	0.8822	0.0275	0.5844	-0.0630	0.5970	0.1257	0.7747	-0.0349	0.7810	0.0653
	D3R	0.8678	-0.0001	0.8774	0.0200	0.5419	-0.0004	0.5622	0.0611	0.7565	-0.0001	0.7666	0.0320
	ATransformer	0.9919	0.0015	0.9905	0.0014	0.9422	0.0011	0.9293	0.0069	0.9805	0.0023	0.9719	0.0013
	DCdetector	0.9930	-0.0001	0.9899	0.0014	0.9556	-0.0006	0.9278	0.0071	0.9875	-0.0001	0.9717	0.0016
	CLAD	0.6670	0.0431	0.5600	0.0005	0.5008	0.0040	0.4334	0.0004	0.5790	0.0126	0.5132	0.0053
	DADA	0.5961	0.0175	0.5963	0.0632	0.5124	0.0499	0.4224	0.0652	0.7628	0.009	0.7681	0.0332
	ReCATS	0.9410	0.0559	0.9279	0.0000	0.8468	0.1040	0.8117	0.0001	0.9383	0.0515	0.9275	0.0000
SWaT	ModernTCN	0.9695	0.0039	0.9382	0.0005	0.8936	0.0491	0.7380	0.0011	0.9333	0.0241	0.8492	0.0006
	D3R	0.9612	-0.0128	0.9704	0.0216	0.8252	-0.1082	0.8888	0.1231	0.9200	-0.0581	0.9483	0.0692
	ATransformer	0.9925	0.0007	0.9903	0.0021	0.9798	0.0024	0.9604	0.0034	0.9933	0.0004	0.9876	0.0025
	DCdetector	0.9914	-0.0020	0.9926	0.0029	0.9788	-0.0025	0.9712	0.0044	0.9927	-0.0023	0.9947	0.0029
	CLAD	0.8321	-0.0027	0.7872	0.0044	0.7443	-0.0015	0.6562	0.0023	0.8139	-0.0038	0.7529	0.0050
	DADA	0.8506	-0.0086	0.8436	0.0333	0.6599	-0.0125	0.5733	0.0389	0.8362	-0.0065	0.8023	0.0222
	ReCATS	0.9878	0.0031	0.9855	0.0075	0.9670	0.0109	0.9762	0.0128	0.9896	0.0017	0.9738	0.0063

Table 2: Average Runtime per Task (seconds).

Methods	Weather	PSM	MSL	NIPS-TS-GECCO	SMAP	SWaT
ATransformer	5.4	23.1	25	22	68.2	76.8
DCdetector	15.3	42.9	55	19	167.2	259.2
D3R	23.4	114.4	189	180	459.8	499.2
ModernTCN	6.3	39.6	11	9	101.2	453.6
CLAD	45.0	209.0	136	37	891	1065.6
ReCATS	<u>6.3</u>	23.1	<u>25</u>	32	57.2	69.6

Table 3: Average Training time per Task (seconds).

Methods	Weather	PSM	MSL	NIPS-TS-GECCO	SMAP	SWaT
ATransformer	3.1	12.1	10.5	12.2	32.2	31.4
DCdetector	3.1	14.6	21.3	9.9	41.7	79
D3R	21.4	98.8	112.7	119.2	276.2	258.3
ModernTCN	1.7	8.2	2.0	4.3	9.0	30.1
ReCATS	5.2	19.5	22.3	25.4	38.3	57.0

D3R [29], Anomaly Transformer [32], DCdetector [33] and DADA [26]. Additionally, we adapt the CLAD Benchmark [4] which is originally designed for pixel level image tasks to serve as a competitive baseline for time series continual learning.

Evaluation Metrics. We evaluate anomaly detection using Accuracy, F1-score, and ROC-AUC. Following prior studies in continual learning [7, 21, 31], we also adopt task-level metrics derived from the performance matrix, including Average Performance (Average), Backward Transfer (BWT) for knowledge retention, Forward

Transfer (FWT) for adaptation to future tasks, and Forgetting Measure (FM) for performance degradation. The detailed calculation formulas for these metrics are provided in Appendix A.2.

Implementation Details. ReCATS is implemented in PyTorch and evaluated on a single NVIDIA RTX 4070 GPU. Training is performed using the Adam optimizer, the learning rate is set to $1e-3$ for the local training phase and $1e-4$ for the global alignment phase. To ensure fair comparison, all baseline methods are evaluated using the same task partitions, windowing strategy, and evaluation protocol. When available, we adopt the official implementations and recommended hyperparameter settings of competing methods.

5.2 Overall Performance Comparison (RQ1)

We evaluate ReCATS against six representative baselines on six real-world multivariate time series datasets (Table 1). ReCATS consistently achieves the highest or near-highest Accuracy, F1-score, and ROC-AUC across most datasets. On Weather, it surpasses DCdetector by over 5% in Accuracy and F1-score, with positive BWT (≈ 0.026) and negligible FM, demonstrating strong retention under non-stationarity. In industrial benchmarks like PSM, it maintains stable high performance (Accuracy ≈ 0.98 – 0.99 ; F1 ≈ 0.96 – 0.98) while mitigating forgetting and enabling forward knowledge transfer. On NIPS-TS-GECCO and SMAP, it improves F1-score by up to 10% over baselines, achieving substantial positive BWT (up to 0.104) and near-zero FM, highlighting its effectiveness in handling heterogeneous operational regimes. On SWaT, ReCATS shows slightly lower relative performance, likely due to the dataset’s severe class imbalance and rapid, transient anomalies, which pose challenges for continual adaptation and anomaly modeling. In contrast, non-continual methods frequently exhibit negative BWT and higher FM, reflecting pronounced performance degradation under evolving distributions.

5.3 Efficiency and Scalability Evaluation (RQ2)

5.3.1 Efficiency Evaluation. We evaluate the efficiency of ReCATS from both runtime and training perspectives. Table 2 reports the average runtime per task, including the complete pipeline from data loading to anomaly discrimination, while Table 3 presents the average training time per task. DADA is excluded due to the unavailability of its pre-training code.

Overall, ReCATS achieves competitive efficiency across all datasets while maintaining superior continual anomaly detection performance. As shown in Table 2, ReCATS consistently outperforms replay-based continual learning methods such as D3R and CLAD in runtime efficiency, particularly on high-dimensional industrial datasets including SMAP and SWaT. For example, on SWaT, ReCATS completes a single task in 69.6 seconds, significantly faster than DCdetector, D3R, and CLAD.

Table 3 further shows that ReCATS maintains moderate training overhead despite employing dual-phase continual optimization and pseudo-sample generation. Although lightweight models such as ModernTCN achieve lower training cost, ReCATS provides a favorable trade-off between computational efficiency and anomaly detection effectiveness. These results demonstrate that ReCATS

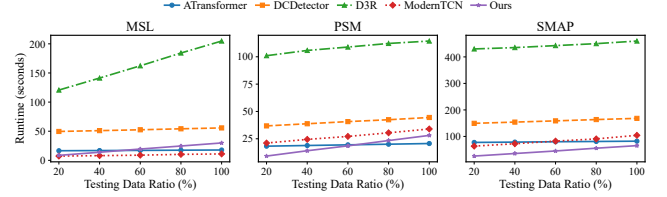


Figure 3: Scalability evaluation

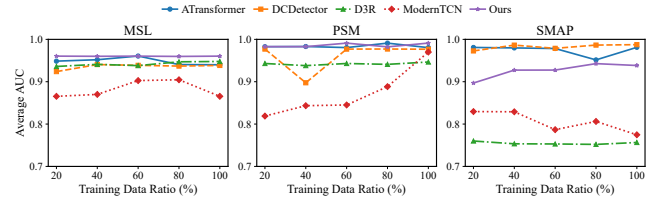


Figure 4: Model generality evaluation

scales effectively to large and high-dimensional streaming environments while remaining practical for continual anomaly detection in real-world industrial systems.

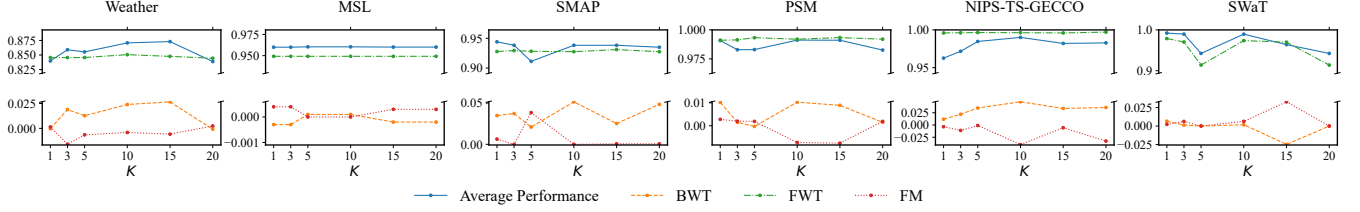
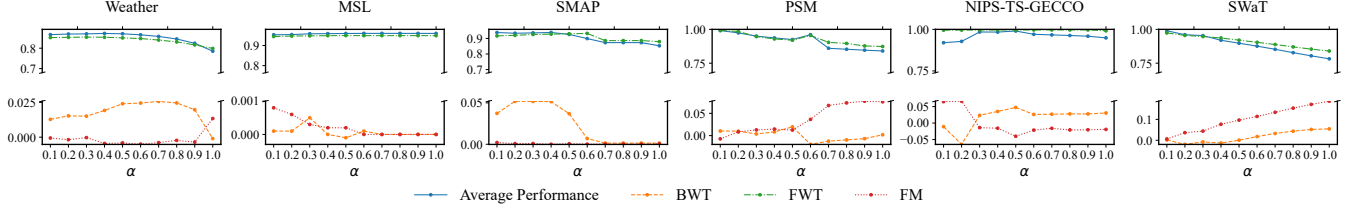
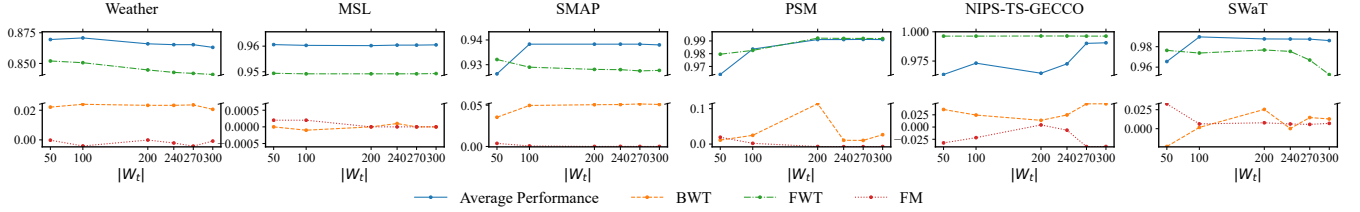
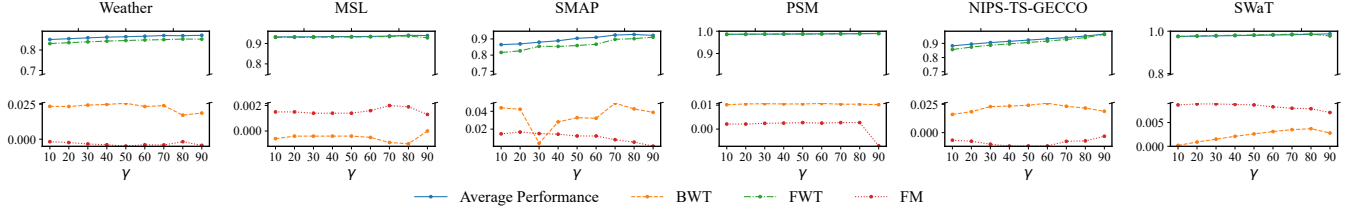
5.3.2 Scalability Evaluation. Figure 3 illustrates the runtime scalability of different methods under varying cardinality of the test set (20%–100%) on three large-scale datasets, namely MSL, PSM, and SMAP. These datasets are selected due to their relatively large data volume, which better reflects scalability behavior in practical settings. CLAD is not included in this comparison, as its overall runtime is approximately 7–8 $\times$ higher than other methods (as shown in Table 2), making its scalability trends less informative for comparison.

As the cardinality of the test set increases, the runtime of all methods grows consistently across datasets, indicating stable scaling behavior. Compared with transformer-based baselines and replay-based methods, ReCATS exhibits a significantly lower runtime growth rate as the test set size increases. In particular, ReCATS maintains near-linear scalability on all three datasets, whereas methods such as D3R incur substantially higher computational overhead under larger test sets. These results demonstrate that ReCATS is computationally efficient and well suited for continual anomaly detection in large-scale scenarios.

5.4 Model Generality Evaluation (RQ3)

We evaluate the model generality of ReCATS by varying the training data ratio (20%, 40%, 60%, 80%, and 100% of the original data) on MSL, PSM, and SMAP using the same baseline methods as in the scalability evaluation. Models trained under different sampling ratios are evaluated on their corresponding test sets, and the Average AUC across all tasks is reported.

As shown in Figure 4, reducing the amount of training data degrades the performance of all methods. Nevertheless, ReCATS maintains substantially more stable performance across different data ratios than the baselines. In particular, under low-data settings (e.g., 20% and 40%), ReCATS consistently achieves higher Average AUC on all three datasets, demonstrating strong robustness to data scarcity. Even with the full dataset, ReCATS remains competitive or superior.

Figure 5: Impact of Parameter K on Detection Results.Figure 6: Impact of Parameter α on Detection Results.Figure 7: Impact of Parameter $|W_t|$ on Detection Results.Figure 8: Impact of Parameter γ on Detection Results.

Overall, these results indicate that ReCATS can effectively leverage limited training data while maintaining reliable continual anomaly detection performance.

5.5 Parameter Sensitivity Analysis (RQ4)

In this section, we examine the sensitivity and robustness of ReCATS with respect to key hyperparameters, using ROC-AUC-based continual learning metrics.

Effect of the Number of Mixture Components K . The parameter K controls the complexity of the latent distribution and affects the model's ability to characterize multi-regime data. As shown in Figure 5, the ROC-AUC-based performance generally improves as K increases from 1 to 10, and then gradually stabilizes or slightly degrades. The best overall performance is achieved at $K = 10$, indicating that moderate mixture components improve heterogeneous distribution modeling, while excessively large K may introduce noise and reduce generalization performance.

Effect of the Threshold Coefficient α . The threshold coefficient α balances the static (global) and adaptive (local) thresholds in anomaly

detection. As shown in Figure 6, the overall performance first improves and then declines as α increases from 0.1 to 1.0. When α becomes excessively large (e.g., $\alpha > 0.7$), over-reliance on the adaptive threshold weakens global statistical constraints, leading to unstable anomaly decisions and degraded performance. This observation highlights the importance of balancing long-term distributional structure with short-term error fluctuations.

Effect of the Time Window Length $|W_t|$. The time window length $|W_t|$ determines the temporal scope for computing local reconstruction statistics and task partitioning. As shown in Figure 7, although the optimal window size varies across datasets, a consistent preference for moderate values can be observed.

Excessively small windows (e.g., 50) fail to capture sufficient temporal context, while overly large windows (e.g., 300) smooth out localized anomalies and reduce detection sensitivity. For example, a window size of 100 achieves the best ROC-AUC on Weather, whereas 270 performs best on both PSM and NIPS-TS-GECCO. These results indicate that an appropriate window size balances temporal dependency modeling and noise suppression.

Table 4: Ablation study results. Bold indicates the best.

Dataset	Methods	Accuracy				F1				ROC_AUC			
		Average	BWT	FWT	FM	Average	BWT	FWT	FM	Average	BWT	FWT	FM
Weather	ReCATS	0.8617	0.0263	0.8356	-0.0053	0.8767	0.0259	0.8530	-0.0054	0.8708	0.0240	0.8506	-0.0041
	w/o CL	0.8258	-0.0008	0.8289	0.0026	0.8406	-0.0007	0.8466	0.0026	0.8397	-0.0009	0.8452	0.0028
	w/o attention	0.8586	0.0211	0.8330	0.0004	0.8738	0.0219	0.8504	0.0001	0.8682	0.0184	0.8484	0.0006
	w/o GM	0.8259	-0.0005	0.8294	0.0014	0.8408	-0.0005	0.8469	0.0013	0.8399	-0.0004	0.8456	0.0014
	w/o DDT	0.8099	0.0022	0.8200	0.0112	0.8496	0.0042	0.8572	0.0089	0.7854	-0.0010	0.7988	0.0134
PSM	ReCATS	0.9930	0.0069	0.9938	-0.0038	0.9831	0.0111	0.9866	-0.0051	0.9912	0.0103	0.9920	-0.0073
	w/o CL	0.9891	0.0001	0.9914	0.0002	0.9783	0.0005	0.9856	0.0003	0.9845	0.0001	0.9900	0.0001
	w/o attention	0.9937	0.0091	0.9923	0.0000	0.9843	0.0146	0.9833	0.0007	0.9917	0.0122	0.9904	0.0004
	w/o GM	0.9923	0.0076	0.9928	0.0031	0.9813	0.0111	0.9844	0.0074	0.9908	0.0102	0.9911	0.0028
	w/o DDT	0.8828	0.0426	0.9027	0.0681	0.7883	-0.0312	0.8284	0.1368	0.8404	0.0019	0.8734	0.0759
SMAP	ReCATS	0.9410	0.0559	0.9279	0.0000	0.8468	0.1040	0.8117	0.0001	0.9383	0.0515	0.9275	0.0000
	w/o CL	0.8869	0.0001	0.9279	0.0000	0.7451	0.0000	0.8117	0.0001	0.8887	0.0000	0.9275	0.0000
	w/o attention	0.9362	0.0501	0.9285	0.0062	0.8346	0.0872	0.8141	0.0139	0.9356	0.0482	0.9278	0.0039
	w/o GM	0.9461	0.0417	0.9284	0.0071	0.8418	0.0617	0.8119	0.0168	0.9441	0.0349	0.9278	0.0063
	w/o DDT	0.8777	0.0023	0.8955	0.0008	0.7063	0.0076	0.7378	0.0007	0.8528	0.0014	0.8794	0.0005
MSL	ReCATS	0.9730	0.0001	0.9658	0.0000	0.9259	0.0002	0.9035	0.0000	0.9604	0.0001	0.9492	0.0000
	w/o CL	0.9730	0.0000	0.9658	0.0000	0.9258	-0.0001	0.9035	0.0001	0.9603	0.0000	0.9492	0.0000
	w/o attention	0.9727	-0.0003	0.9658	0.0004	0.9251	-0.0006	0.9034	0.0009	0.9601	-0.0002	0.9492	0.0003
	w/o GM	0.9725	-0.0005	0.9657	0.0005	0.9248	-0.0009	0.9033	0.0012	0.9600	-0.0003	0.9492	0.0004
	w/o DDT	0.9728	0.0000	0.9656	0.0000	0.9254	0.0000	0.9030	0.0000	0.9602	0.0000	0.9491	0.0000
NIPS-TS-GECCO	ReCATS	0.9947	0.0178	0.9957	-0.0114	0.8835	0.0850	0.9574	-0.0464	0.9901	0.0474	0.9963	-0.0400
	w/o CL	0.9874	0.0000	0.9973	0.0000	0.9416	0.0000	0.9759	0.0000	0.9630	0.0000	0.9971	0.0000
	w/o attention	0.9874	0.0000	0.9973	0.0000	0.9416	0.0000	0.9759	0.0000	0.9630	0.0000	0.9971	0.0000
	w/o GM	0.9730	0.0045	0.9952	-0.0012	0.8326	0.0244	0.9536	-0.0032	0.9626	0.0119	0.9960	-0.0035
	w/o DDT	0.9324	0.0235	0.9858	-0.0071	0.7754	0.1092	0.8597	-0.0289	0.9489	0.0306	0.9912	-0.0192
SWaT	ReCATS	0.9878	0.0031	0.9855	0.0075	0.9670	0.0109	0.9762	0.0128	0.9896	0.0017	0.9738	0.0063
	w/o CL	0.8869	0.0000	0.9272	0.0000	0.7451	0.0000	0.8117	0.0000	0.8887	0.0000	0.9275	0.0000
	w/o attention	0.9868	0.0039	0.9854	0.0078	0.9641	0.0121	0.9303	0.0151	0.9889	0.0023	0.9739	0.0064
	w/o GM	0.9911	0.0089	0.9841	0.0028	0.9744	0.0206	0.9354	0.0045	0.9924	0.0065	0.9789	0.0023
	w/o DDT	0.7783	0.1343	0.8574	0.1720	0.6886	0.0652	0.6898	0.2156	0.7824	0.0545	0.8399	0.1870

Effect of the Quantile Threshold Parameter γ . The parameter γ controls the quantile used for determining the static threshold in the DDT module, thereby affecting anomaly discrimination sensitivity. As shown in Figure 8, the ROC-AUC-based performance remains relatively stable when γ varies from 10 to 90.

5.6 Ablation Study (RQ5)

We conduct an ablation study to evaluate the contributions of the key components in ReCATS, including the continual learning mechanism (CL), the attention-based prior generator, the Gaussian mixture (GM) latent modeling, and the dynamic dual-scale thresholding (DDT) strategy. The results are reported in Table 4.

Overall, the complete ReCATS framework achieves the best or consistently strong performance across most datasets and metrics, validating the effectiveness of jointly integrating these components.

Removing the continual learning mechanism (*w/o CL*) significantly degrades BWT and FWT, especially on Weather, SMAP, and NIPS-TS-GECCO, demonstrating its importance in mitigating catastrophic forgetting. Excluding the attention-based prior generator (*w/o attention*) reduces Accuracy and F1-score on datasets with strong temporal non-stationarity, highlighting the benefit of contextual temporal modeling. Removing Gaussian mixture latent modeling (*w/o GM*) leads to notable declines in F1-score and ROC-AUC, indicating that unimodal latent representations are insufficient for complex distributions. In addition, removing the dynamic dual-scale thresholding strategy (*w/o DDT*) causes consistent performance drops, showing that fixed thresholds are less robust under distribution shifts.

These results demonstrate that the four components are complementary: continual learning improves long-term stability, attention enhances temporal adaptability, Gaussian mixture modeling increases representation expressiveness, and DDT enables robust anomaly decision-making under evolving distributions.

6 Conclusion

We present ReCATS, a replay-free continual anomaly detection framework for multivariate time series. By combining a dual-phase training strategy with dynamic dual-scale thresholding, ReCATS effectively balances stability and plasticity while addressing decision boundary drift under non-stationary data distributions. Experiments on six benchmark datasets demonstrate that ReCATS consistently outperforms existing methods in both detection accuracy and knowledge retention. Moreover, we provide the first systematic evaluation of continual time-series anomaly detection using standard continual learning metrics, establishing a reproducible baseline for future research. Although ReCATS demonstrates strong effectiveness under sequential continual learning settings, the current framework assumes predefined task boundaries. Extending ReCATS to boundary-free or drift-triggered streaming scenarios remains an important direction for future work.

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References

- [1] Ahmed Abdulaal, Zhuanghua Liu, and Tomer Lincewicz. 2021. Practical Approach to Asynchronous Multivariate Time Series Anomaly Detection and Localization. In *KDD*. 2485–2494.
- [2] Archana Anandakrishnan, Senthil Kumar, Alexander R. Statnikov, Tanveer A. Faruque, and Di Xu. 2017. Anomaly Detection in Finance: Editors' Introduction. In *KDD*, Vol. 71. 1–7.
- [3] Markus M. Breunig, Hans-Peter Kriegel, Raymond T. Ng, and Jörg Sander. 2000. LOF: Identifying Density-Based Local Outliers. In *SIGMOD*. 93–104.
- [4] Nikola Bugarin, Jovana Bugaric, Manuel Barusco, Davide Dalle Pezze, and Gian Antonio Susto. 2024. Unveiling the Anomalies in an Ever-Changing World: A Benchmark for Pixel-Level Anomaly Detection in Continual Learning. In *CVPR*. 4065–4074.
- [5] Andrew A. Cook, Goksel Misirli, and Zhong Fan. 2020. Anomaly Detection for IoT Time-Series Data: A Survey. *IoT-J* 7, 7 (2020), 6481–6494.
- [6] Ryan Elwell and Robi Polikar. 2011. Incremental Learning of Concept Drift in Nonstationary Environments. *IEEE Trans. Neural Netw.* 22, 10 (2011), 1517–1531.
- [7] Kamil Faber, Roberto Corizzo, Bartłomiej Sniezynski, and Nathalie Japkowicz. 2024. Lifelong Continual Learning for Anomaly Detection: New Challenges, Perspectives, and Insights. *IEEE Access* 12 (2024), 41364–41380.
- [8] Koosha Golmohammadi and Osmar R. Zaiane. 2015. Time series contextual anomaly detection for detecting market manipulation in stock market. In *DSAA*. 1–10.
- [9] Gastón García González, Pedro Casas, and Alicia Fernández. 2023. Fake it till you Detect it: Continual Anomaly Detection in Multivariate Time-Series using Generative AI. In *EuroS&P*. 558–566.
- [10] Kyle Hundman, Valentino Constantinou, Christopher Laporte, Ian Colwell, and Tom Söderström. 2018. Detecting Spacecraft Anomalies Using LSTMs and Non-parametric Dynamic Thresholding. In *KDD*. 387–395.
- [11] Neha Kant and Manish Mahajan. 2018. Time-series outlier detection using enhanced k-means in combination with pso algorithm. In *ICoEVC*. 363–373.
- [12] James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A. Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, Demis Hassabis, Claudia Clopath, Dharshan Kumaran, and Raia Hadsell. 2017. Overcoming catastrophic forgetting in neural networks. *PNAS* 114, 13 (2017), 3521–3526.
- [13] Matthias De Lange, Rahaf Aljundi, Marc Masana, Sarah Parisot, Xu Jia, Ales Leonardis, Gregory G. Slabaugh, and Tinne Tuytelaars. 2022. A Continual Learning Survey: Defying Forgetting in Classification Tasks. *TPAMI* 44, 7 (2022), 3366–3385.
- [14] Fei Tony Liu, Kai Ming Ting, and Zhi-Hua Zhou. 2012. Isolation-Based Anomaly Detection. *TKDD* 6, 1 (2012), 3:1–3:39.
- [15] Donghao Luo and Xue Wang. 2024. ModernTCN: A Modern Pure Convolution Structure for General Time Series Analysis. In *ICLR*.
- [16] Qian Ma, Qiuyang Li, Xingshun Liu, Toufiq Imran Khoshnobia, Mei Bai, Xite Wang, Bo Ning, and Shikai Guo. 2025. NADHS: Online Anomaly Detection for High-dimensional Data Streams. *IEEE Transactions on Instrumentation and Measurement* (2025).
- [17] Aditya P. Mathur and Nils Ole Tippenhauer. 2016. SWaT: a water treatment testbed for research and training on ICS security. In *CySWater*. 31–36.
- [18] Steffen Moritz, Frederik Rehbach, Sowmya Chandrasekaran, Margarita Rebollo, and Thomas Bartz-Beielstein. 2018. GECCO Industrial Challenge 2018 Dataset: A water quality dataset for the “Internet of Things: Online Anomaly Detection for Drinking Water Quality” competition. In *GECCO*.
- [19] Guansong Pang, Chunhua Shen, Longbing Cao, and Anton Van Den Hengel. 2021. Deep Learning for Anomaly Detection: A Review. *CSUR* 54, 2 (2021).
- [20] Daehyung Park, Yuuna Hoshi, and Charles C. Kemp. 2018. A Multimodal Anomaly Detector for Robot-Assisted Feeding Using an LSTM-Based Variational Autoencoder. *RAL* 3, 2 (2018), 1544–1551.
- [21] Natalia Díaz Rodríguez, Vincenzo Lomonaco, David Filliat, and Davide Maltoni. 2018. Don't forget, there is more than forgetting: new metrics for Continual Learning. *CoRR* abs/1810.13166 (2018).
- [22] Mayu Sakurada and Takehisa Yairi. 2014. Anomaly Detection Using Autoencoders with Nonlinear Dimensionality Reduction. In *MLSDA*. 4–11.
- [23] Sebastian Schmidl, Phillip Wenig, and Thorsten Papenbrock. 2022. Anomaly Detection in Time Series: A Comprehensive Evaluation. *PVLDB* 15, 9 (2022), 1779–1797.
- [24] Bernhard Schölkopf, Robert C. Williamson, Alexander J. Smola, John Shawe-Taylor, and John C. Platt. 1999. Support Vector Method for Novelty Detection. In *NeurIPS*. 582–588.
- [25] Lifeng Shen, Zhuocong Li, and James T. Kwok. 2020. Timeseries Anomaly Detection using Temporal Hierarchical One-Class Network. In *NeurIPS*.
- [26] Qichao Shentu, Beibu Li, Kai Zhao, Yang Shu, Zhongwen Rao, Lujia Pan, Bin Yang, and Chenjuan Guo. 2025. Towards a General Time Series Anomaly Detector with Adaptive Bottlenecks and Dual Adversarial Decoders. In *ICLR*.
- [27] Jie Song, Sijia Zhang, Junghoon Park, Yu Gu, and Ge Yu. 2026. Dimension Expansion for Learning Money Laundering Activities Hidden in Transaction Network. *IEEE Trans. Comput. Soc. Syst.* 13, 1 (2026), 251–262.
- [28] Ya Su, Youjian Zhao, Chenhao Niu, Rong Liu, Wei Sun, and Dan Pei. 2019. Robust Anomaly Detection for Multivariate Time Series through Stochastic Recurrent Neural Network. In *SIGKDD*. 2828–2837.
- [29] Chengsen Wang, Zirui Zhuang, Qi Qi, Jingyu Wang, Xingyu Wang, Haifeng Sun, and Jianxin Liao. 2023. Drift doesn't Matter: Dynamic Decomposition with Diffusion Reconstruction for Unstable Multivariate Time Series Anomaly Detection. In *NeurIPS*. 10758–10774.
- [30] Jianing Wang, Zheng Hua, Wan Zhang, Shengjia Hao, Yuqiong Yao, and Maoguo Gong. 2025. CL-BioGAN: Biologically Inspired Cross-Domain Continual Learning for Hyperspectral Anomaly Detection. *TGRS* 63 (2025), 1–15.
- [31] Liyuan Wang, Xingxing Zhang, Hang Su, and Jun Zhu. 2024. A Comprehensive Survey of Continual Learning: Theory, Method and Application. *PAMI* 46, 8 (2024), 5362–5383.
- [32] Jiehui Xu, Haixu Wu, Jianmin Wang, and Mingsheng Long. 2022. Anomaly Transformer: Time Series Anomaly Detection with Association Discrepancy. In *ICLR*.
- [33] Yiyuan Yang, Chaoli Zhang, Tian Zhou, Qingsong Wen, and Liang Sun. 2023. DCdetector: Dual Attention Contrastive Representation Learning for Time Series Anomaly Detection. In *KDD*. 3033–3045.
- [34] Zahra Zamanzadeh Darban, Geoffrey I. Webb, Shirui Pan, Charu Aggarwal, and Mahsa Salehi. 2024. Deep Learning for Time Series Anomaly Detection: A Survey. *CSUR* 57, 1 (2024).
- [35] Friedemann Zenke, Ben Poole, and Surya Ganguli. 2017. Continual Learning Through Synaptic Intelligence. In *ICML*, Vol. 70. 3987–3995.
- [36] Bin Zhou, Shenghua Liu, Bryan Hooi, Xueqi Cheng, and Jing Ye. 2019. BeatGAN: Anomalous Rhythm Detection using Adversarially Generated Time Series. In *IJCAI*. 4433–4439.

A Details of Experimental Settings

A.1 Datasets

Table 5: General information about six datasets.

Dataset	Source	Dimension	#Tasks
SMAP	NASA Space Sensors	25	22
PSM	eBay Server Machine	25	11
Weather	NOAA Weather Station	8	9
MSL	NASA Space Sensors	55	10
NIPS-TS-GECCO	Water Quality for IoT	9	10
SWaT	Infrastructure System	51	24

Detailed dataset descriptions. We evaluate ReCATS on six publicly available benchmark datasets spanning diverse application domains, including environmental monitoring, server systems, space telemetry, and critical infrastructure. These datasets differ substantially in dimensionality, data sources, and operational contexts, providing a comprehensive testbed for evaluating continual anomaly detection under non-stationary conditions.

Specifically, Weather records meteorological observations with eight sensor variables, while PSM is a 25-dimensional dataset collected from server machines. MSL and SMAP, provided by NASA, contain telemetry data from space missions, including rover operations and soil moisture measurements. In addition, SWaT and NIPS-TS-GECCO represent sensor-based datasets from critical infrastructure and industrial control systems. Following the continual learning setting, each dataset is chronologically partitioned into multiple consecutive task segments to simulate evolving data distributions over time.

- **Weather** records meteorological observations from Offutt Air Force Base, including 8 variables such as temperature, air pressure, and wind speed.
- **PSM** is a 25-dimensional dataset collected from server machines at eBay, reflecting real-world system monitoring scenarios.
- **MSL**, provided by NASA, contains 55-dimensional telemetry data from sensors and actuators onboard the Curiosity Mars rover.
- **SMAP**, also collected by NASA, provides Soil Moisture Active Passive measurements and related telemetry data, comprising 25 environmental variables.
- **SWaT** is a 51-dimensional sensor dataset collected from a water treatment testbed operating under continuous industrial control conditions.
- **NIPS-TS-GECCO** is an Internet-of-Things dataset for drinking water quality monitoring, consisting of 9-dimensional multivariate time series data.

These datasets exhibit diverse characteristics in terms of dimensionality, sampling frequency, and application background, enabling a comprehensive evaluation of the model’s adaptability across heterogeneous environments.

Continual Task Construction. In continual time-series anomaly detection, each task should correspond to a coherent temporal segment and include both training and testing data. However, most benchmark datasets provide training splits containing only normal samples, which makes direct task-level partitioning infeasible.

To address this limitation, we construct tasks using the original test split, which is sufficiently long and contains both normal and anomalous samples. For all datasets except Weather, which does

not provide an official train–test split, the original test data are re-partitioned into new training and testing sets. Specifically, normal samples are split into training and testing subsets with a ratio of 7:3, while all anomalous samples are exclusively assigned to the test set. This protocol ensures that training data remain anomaly-free while enabling reliable evaluation within each task.

Following this procedure, the entire time series is chronologically divided into multiple consecutive task segments. The resulting number of tasks for each dataset is summarized in Table 5.

A.2 Evaluation Metrics.

- **Average Performance (Average):** Higher values indicate better performance.

$$\text{Average}(A) = \frac{1}{M} \sum_{j=1}^M A_{M,j} \quad (10)$$

- **Backward Transfer (BWT):** A positive value indicates that knowledge of previous tasks has been retained.

$$\text{BWT}(A) = \frac{1}{M-1} \sum_{j=1}^{M-1} (A_{M,j} - A_{j,j}) \quad (11)$$

- **Forward Transfer (FWT):** This metric quantifies the model’s adaptability to future tasks.

$$\text{FWT}(A) = \frac{M(M-1)}{2} \sum_{i=1}^M \sum_{j=i}^M A_{i,j} \quad (12)$$

- **Forgetting Measure (FM):** This indicator measures the gap between the best performance and the final performance of a task j during the training process. A smaller value indicates better stability and less forgetting.

$$\text{FM}(A) = \frac{1}{M-1} \sum_{j=1}^{M-1} \max_{i \in \{1, \dots, M-1\}} (A_{i,j} - A_{M,j}) \quad (13)$$

		Test Set			
		Task 1	Task 2		Task M
Train Set	Task 1	$A_{1,1}$	$A_{1,2}$		$A_{1,M}$
	Task 2	$A_{2,1}$	$A_{2,2}$		$A_{2,M}$
					
	Task M	$A_{M,1}$	$A_{M,2}$		$A_{M,M}$

Figure 9: Performance Evaluation Matrix.

Taking Accuracy as an example, as shown in Figure 9, the diagonal elements $A_{i,i}$ of matrix A represent the immediate performance. The upper-right section ($i < j$) reflects Forward Transfer (FWT), while the lower-left section ($i > j$) reflects Backward Transfer (BWT). The final row shows the anomaly detection results on all test set tasks after the completion of training on all tasks, which is used to calculate the long-term monitoring performance and the Forgetting Measure (FM) [7, 21, 31].

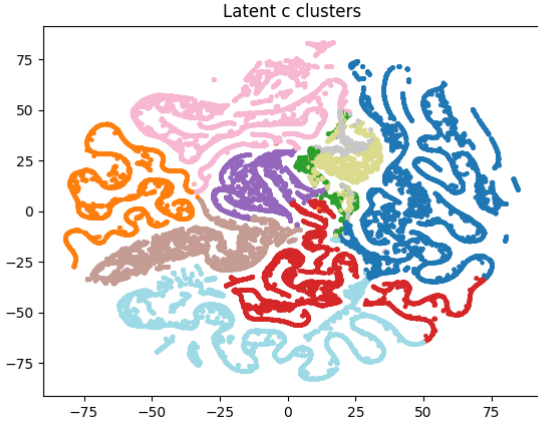


Figure 10: t-SNE visualization of latent variable c on the PSM dataset.

B Additional Analysis of Latent Variable Factorization

The factorization of the latent variables (z, c, w) is motivated by the probabilistic generative modeling of multi-modal time-series data. Real-world multivariate time series often contain multiple operational regimes with heterogeneous distributions. To capture such characteristics, ReCATS adopts a Gaussian Mixture VAE framework, where the discrete latent variable c represents mixture component assignments, the continuous variable w parameterizes prior distribution shifts, and the latent representation z models intra-mode temporal variations. This factorization enables the model to separately capture inter-mode and intra-mode variations that are difficult to represent using a single latent variable.

The disentanglement is induced by the model structure. Specifically, c determines the mixture component assignment, $z \sim p(z|c, w)$ is conditioned on the selected component, and w further shapes the latent prior distribution. As a result, these variables play complementary and non-redundant roles in modeling evolving time-series distributions.

To validate the necessity of this factorization, we further conduct ablation experiments on the Weather dataset using ROC-AUC-based continual learning metrics. The results are shown in Table 6.

Table 6: Ablation analysis of latent variable factorization on the Weather dataset.

Method	Average ROC_AUC	BWT	FWT	FM
ReCATS	0.8708	0.0240	0.8506	-0.0041
w/o w	0.8624	-0.0010	0.8669	0.0144
w/o c	0.8663	0.0190	0.8524	0.0034
w/o c & w	0.8662	-0.0018	0.8738	0.0153

Removing either c or w consistently degrades continual learning performance, particularly in BWT and FM, indicating weaker knowledge retention and increased forgetting. Removing both variables further destabilizes the latent representation structure. These results demonstrate that the factorized latent design effectively improves distribution modeling and continual adaptation under non-stationary environments.

To further evaluate the effectiveness of the discrete latent variable c , we additionally conduct clustering and t-SNE visualization analysis on the learned latent representations. The visualization results show compact and well-separated clusters corresponding to different operational regimes, indicating that c effectively captures the underlying mode structure of non-stationary multivariate time series. Figure 10 presents the visualization results using the PSM dataset as an example.

C Additional Evaluation without Point Adjustment

Following prior works, we adopt Point Adjustment (PA) as the standard evaluation protocol. As discussed in ATransformer, PA is meaningful for real-world anomaly detection because anomalies are often event-level, where detecting any point within an anomalous segment is sufficient for practical inspection.

To further evaluate robustness, we additionally report results under the No-PA setting on the PSM dataset. The results are shown in Table 7.

Table 7: Performance comparison under the No-PA setting on the PSM dataset.

Method	Average Accuracy	Average F1-score	Average ROC-AUC
D3R	0.3719	0.4165	0.5141
ATransformer	0.5368	0.0190	0.4999
ModernTCN	0.5502	0.0584	0.5146
DADA	0.6195	0.4853	0.5743
ReCATS	0.6152	0.5882	0.5788

Although the performance of all methods decreases without PA, ReCATS consistently achieves the best F1-score and ROC-AUC, demonstrating superior robustness under stricter anomaly evaluation settings.